

Technical Report – 2Market Analysis

Introduction

2Market is an international grocery retailer that operates both online and through physical stores. To allow the business to stay competitive, grow their market share the company will need to analyse the relationship between their customers' demographics and buying habits along with the interactions via the different marketing channels. This analysis will ascertain whether buying habits are associated with demographic factors such as marital status or location, as well as identify patterns between purchases and interactions with the company's marketing campaigns. Insights gained from this analysis will help determine which are the most successful marketing strategies to pursue.

Data Cleaning within Excel

There are two csv files of raw data, along with a text file containing the meta data as a description, to begin with. Within the first csv file, marketing_data.csv, there were customer demographics including birth year, marital status, and country as well as a spend of customers purchasing by different categories. The demographics and spending patterns warrant further exploration. The raw data within the csv file was accessed through excel and data cleaning was commenced. Any blank spaces or cells were removed; this was to ensure that the data analysed was specific rather than varied. An age column was derived from the birth year data by subtracting the column from 2026, allowing for a better contextual understanding of the customers. An XLOOKUP table was created to standardise marital status (see figure 1). This is so there were fields that had comparative context and therefore could be grouped together.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Marital Status	Clean	Marital Clean										
2	Divorced	Divorced	Divorced										
3	Single	Single	Single										
4	Married	Partner	Partner										
5	Together	Partner	Partner										
6	Single	Single	Single										
7	Single	Single	Single										
8	Married	Partner	Partner										
9	Together	Partner	Partner										
10	Married	Partner	Partner										
11	Married	Partner	Partner					Unique	Raw	Clean			
12	Married	Partner	Partner					Divorced	Divorced	Divorced			
13	Together	Partner	Partner					Single	Single	Single			
14	Widow	Widow	Widow					Married	Married	Partner			
15	Married	Partner	Partner					Together	Together	Partner			
16	Married	Partner	Partner					Widow	Widow	Widow			
17	Married	Partner	Partner					YOLO	YOLO	NA			
18	Married	Partner	Partner					Alone	Alone	Single			
19	Divorced	Divorced	Divorced					Absurd	Absurd	NA			
20	Single	Single	Single										
21	Single	Single	Single										
22	Together	Partner	Partner										
23	Single	Single	Single										
24	Married	Partner	Partner										

Figure 1 - A snapshot of the XLOOKUP table and worksheet for Marital Status

Duplicate data was investigated. It could be possible that a customer's information may have been inputted into the data set more than once, resulting in more than one unique ID. When the duplicates were assessed with customer ID involved, no such duplicates were found. 47 duplicates were removed when the data was assessed without the unique customer ID. The country data was converted from initials to full name (see figure 2). This was to avoid confusion between country initials and state initials.

Exploratory Analysis within Excel

A distribution of age by income revealed some significant outliers. One customer was receiving a salary of \$666,666 and there were three customers that were over the age of 125 (see figure 3). This was a limitation of the data and could impact the analysis, however, they were not removed from the data. This was considered during further analysis.

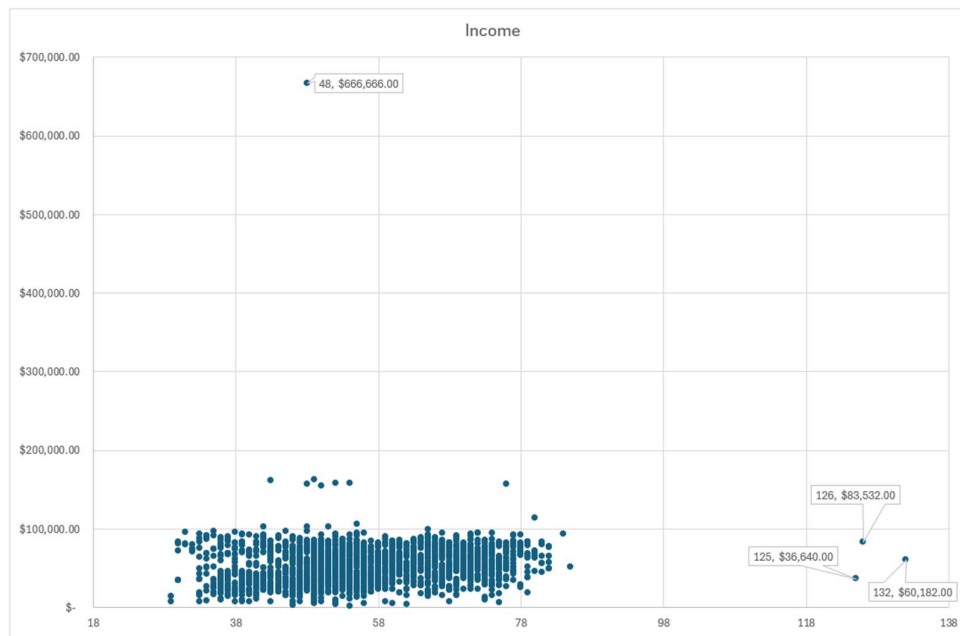


Figure 2 - distribution of ages compared to income with notes next to marks considered outliers

Analysis from pivot tables revealed that the category with the highest spend was alcohol when compared to country and marital status (see figures 4 and 5).

Data Analysis within SQL

To analyse the two data sets simultaneously, they were imported into a PostgreSQL database. Two tables were created using pgAdmin, defining the columns and data types using the query tool (see figures 6 and 7).

The success of different social media adverts compared to country was queried (see figure 8). To achieve this; a sum of the social media adverts data was selected alongside the country column. The sum function was selected because the data was in 1s and 0s, thus making the sum of all the data in the column a total of successful social media

adverts. Because the country data and the social media data were on two different entities, an inner join was selected along the customer ID as a common field to assess all the information that appears in both tables, and ensure only records present in both datasets were included. The customer ID is a common field as they are primary keys within both data sets.

```
87 -- Which social media platform is the most effective method of advertising in each country?
88
89 SELECT
90     md.Country,
91     SUM(ad.Twitter_ad) AS "Twitter Total",
92     SUM(ad.Facebook_ad) AS "Facebook Total",
93     SUM(ad.Instagram_ad) AS "Instagram Total"
94 FROM public.marketing_data md
95 JOIN ad_data ad
96 ON md.ID = ad.ID
97 GROUP BY Country;
```

Figure 3 - pgAdmin query script to ascertain the most effective social media platform in each country

Data Preparation within Tableau

The cleaned marketing data worksheet was saved in a separate excel file and was loaded on to Tableau and a left join with ad_data.csv created. The left join was to allow a link between the two data sets, but primarily focusing on the marketing data. The spending by category was in a wide format which meant that further cleaning within Tableau was required. The spending category columns were pivoted to create a long format data set (see figure 10).

Abc Pivot	# Pivot
Spend Category	Total Spend
Spent on Alcohol	189
Spent on Fish	111
Spent on Meat	379
Spent on chocolates	189
Spent on vegetables	104
Spent On Commodities	218
Spent on Alcohol	464

Figure 4 - a sample of the pivot table. The column head detailing pivot

There were also calculated fields created to analyse the advertising data. Using the 0s and 1s within the data, calculations could be made compared to campaign success

between the two data sets by using IF functions (see figure 11). This allowed for a breakdown of different advertising channels to be analysed in one succinct graph.

Data visualisation within Tableau

The dashboard has KPIs stating headline figures that should stand out to the stakeholders (see figure 12). There are four bar charts that show quantifiable measures of the data. They have been used to allow comparisons between different categories. The charts are in a 'Z' formation in regard to importance when viewing the dashboard; along with using Tableau's colour-blind palette, a legend on the side, makes it accessible to every stakeholder using it. In order to achieve a more granular analysis of the data; there are filters for some of the charts which detail marital status and country. There is also a filter that affects three of the four charts allowing the user to analyse the success of individual advertising streams.

Insights and Recommendations

From the analysis undertaken, it can be noted that 2Market has a strong brand within Spain. This shows that the marketing campaigns have been successful within this country. South Africa has significant potential for growth, as there has been numerous successful adverts in the country. More marketing within South Africa is recommended to capitalise on growth potential. The analysis has also shown that the bulkmail advertising channel has significant success across the board – this resulted in the most spending when a customer interacted with the advertisement. Although there were fewer brochure advertising responses compared to any others, there was a higher success rate for them – 25%. This is something to keep in mind when moving forward with the campaign. When focusing solely on social media advertising streams, Facebook and Instagram should be prioritised over Twitter due to its higher effectiveness with regards to customer purchasing. Targeting customers in the "Partner" demographic may increase campaign effectiveness as this group demonstrates higher engagement and likely higher household spending. The data does not necessarily show the success of adverts with the customers that are not signed up to the 2Market's database, which can be considered a limitation, and therefore a strong relationship between purchasing behaviour and the influence of advertising cannot be ascertained. Nevertheless, some recommendations can be made. Focusing on single people in South Africa through Facebook can be seen as a successful route for growth. Future marketing through bulkmail can allow for the brand to grow, as this results in the most direct sales.

Appendices

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Marital Status	Clean	Marital Clean										
2	Divorced	Divorced	Divorced										
3	Single	Single	Single										
4	Married	Partner	Partner										
5	Together	Partner	Partner										
6	Single	Single	Single										
7	Single	Single	Single										
8	Married	Partner	Partner										
9	Together	Partner	Partner										
10	Married	Partner	Partner										
11	Married	Partner	Partner					Unique	Raw	Clean			
12	Married	Partner	Partner					Divorced	Divorced	Divorced			
13	Together	Partner	Partner					Single	Single	Single			
14	Widow	Widow	Widow					Married	Married	Partner			
15	Married	Partner	Partner					Together	Together	Partner			
16	Married	Partner	Partner					Widow	Widow	Widow			
17	Married	Partner	Partner					YOLO	YOLO	NA			
18	Married	Partner	Partner					Alone	Alone	Single			
19	Divorced	Divorced	Divorced					Absurd	Absurd	NA			
20	Single	Single	Single										
21	Single	Single	Single										
22	Together	Partner	Partner										
23	Single	Single	Single										
24	Married	Partner	Partner										

Figure 5 - A snapshot of the XLOOKUP table and worksheet for Marital Status

	A	B	C	D	E	F	G	H
1	Country Initial	xlookup country	Clean Country Full					
2	SP	Spain	Spain					
3	CA	Canada	Canada					
4	US	United States of America	United States of America					
5	AUS	Australia	Australia					
6	SP	Spain	Spain			unique	Raw	Clean
7	SP	Spain	Spain			SP	SP	Spain
8	GER	Germany	Germany			CA	CA	Canada
9	SP	Spain	Spain			US	US	United States of America
0	US	United States of America	United States of America			AUS	AUS	Australia
1	IND	India	India			GER	GER	Germany
2	US	United States of America	United States of America			IND	IND	India
3	SP	Spain	Spain			SA	SA	South Africa
4	IND	India	India			ME	ME	Montenegro
5	CA	Canada	Canada					
6	SP	Spain	Spain					
7	SP	Spain	Spain					
8	SP	Spain	Spain					
9	SP	Spain	Spain					
0	CA	Canada	Canada					
1	CA	Canada	Canada					
2	SA	South Africa	South Africa					

Figure 6 - XLOOKUP to create a cleaner country column

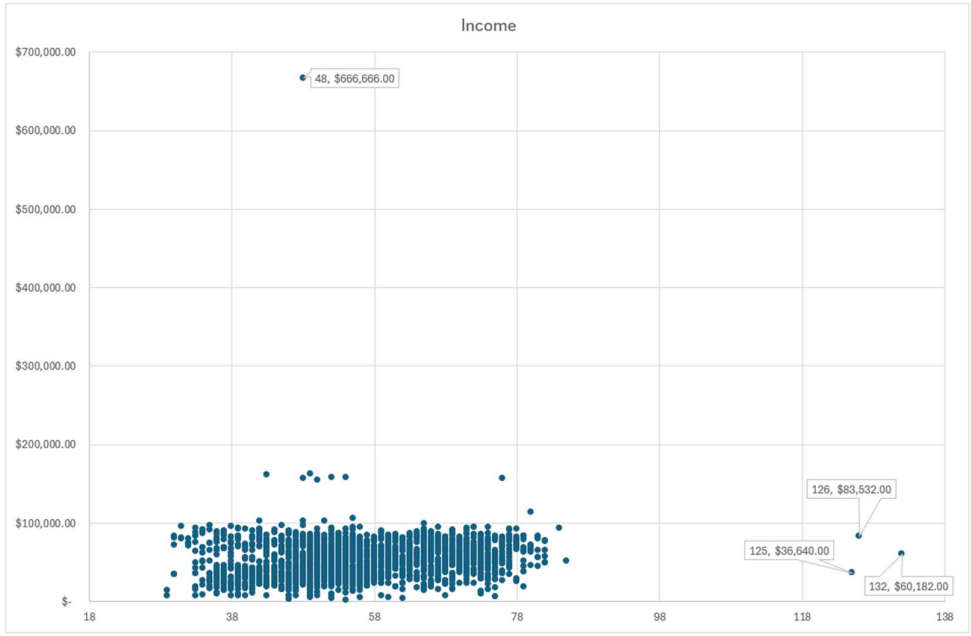


Figure 7 - distribution of ages compared to income with notes next to marks considered outliers

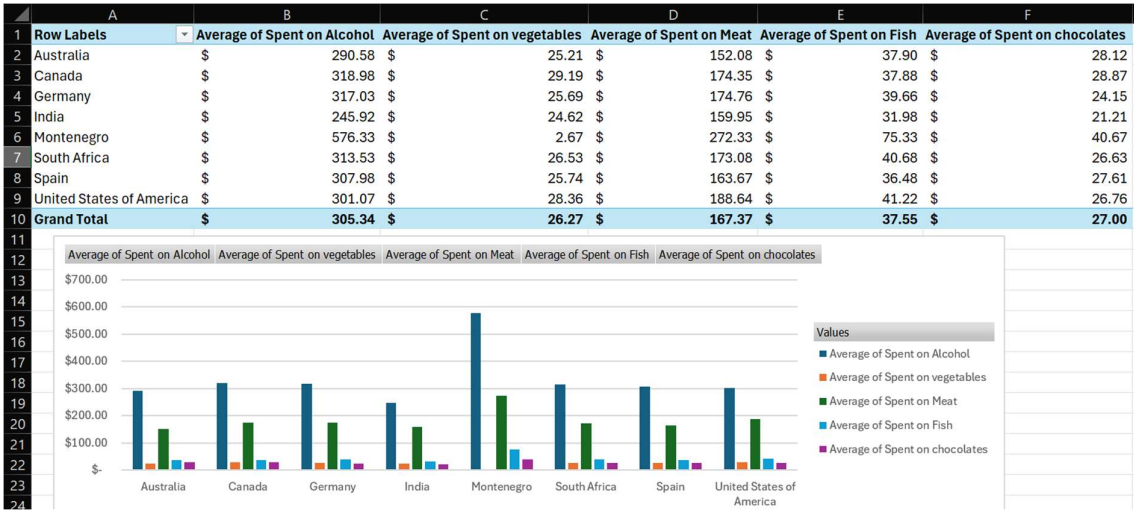


Figure 8 - Pivot table and accompanying graph to show average spend of category compared to country

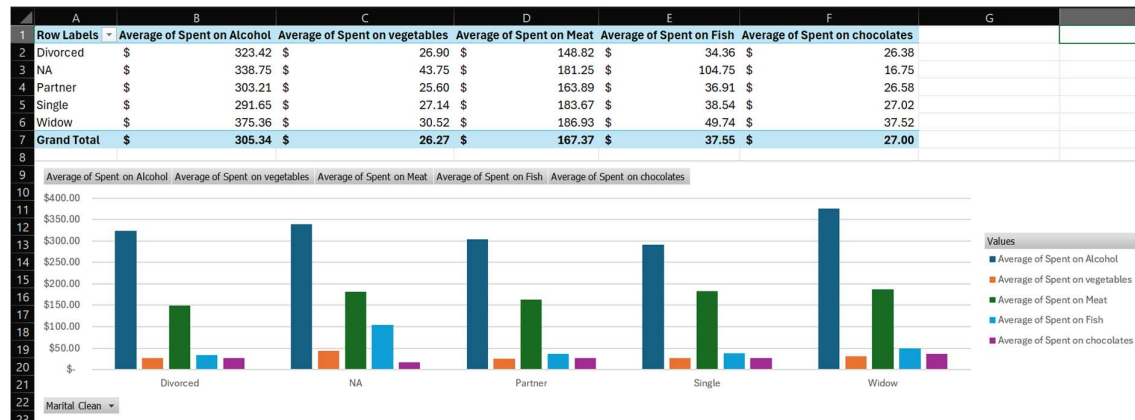


Figure 9 - Pivot table and accompanying graph to show average spend of category compared to marital status

```

1  -- Create marketing_data.csv table
2  CREATE TABLE marketing_data (
3      ID BIGSERIAL PRIMARY KEY,
4      Year_Birth INTEGER,
5      Education TEXT,
6      Marital_Status TEXT,
7      INCOME TEXT,
8      Kidhome INTEGER,
9      Teenhome INTEGER,
10     Dt_Customer TEXT,
11     Recency INTEGER,
12     AmtLiq INTEGER,
13     AmtVeg INTEGER,
14     AmtNonVeg INTEGER,
15     AmtPes INTEGER,
16     AmtChocolates INTEGER,
17     AmtComm INTEGER,
18     NumDeals INTEGER,
19     NumWebBuy INTEGER,
20     NumWalkinBuy INTEGER,
21     NumVisits INTEGER,
22     Response INTEGER,
23     Complain INTEGER,
24     Country VARCHAR (4),
25     Count_success INTEGER);

```

Figure 10 - pgAdmin query script to create the "marketing_data" table

```

76  -- Create ad_data.csv table
77  CREATE TABLE ad_data (
78      ID BIGSERIAL PRIMARY KEY,
79      Bulkmail_ad INTEGER,
80      Twitter_ad INTEGER,
81      Instagram_ad INTEGER,
82      Facebook_ad INTEGER,
83      Brochure_ad INTEGER);

```

Figure 11 - pgAdmin query script to create the "ad_data" table

```

87  -- Which social media platform is the most effective method of advertising in each country?
88
89  SELECT
90      md.Country,
91      SUM(ad.Twitter_ad) AS "Twitter Total",
92      SUM(ad.Facebook_ad) AS "Facebook Total",
93      SUM(ad.Instagram_ad) AS "Instagram Total"
94  FROM public.marketing_data md
95  JOIN ad_data ad
96  ON md.ID = ad.ID
97  GROUP BY Country;

```

Figure 12 - pgAdmin query script to ascertain the most effective social media platform in each country

	country character varying (4) 🔒	Twitter Total bigint 🔒	Facebook Total bigint 🔒	Instagram Total bigint 🔒
1	SP	87	76	89
2	CA	24	18	21
3	IND	10	7	6
4	AUS	6	7	12
5	US	6	7	5
6	ME	0	0	0
7	SA	20	20	21
8	GER	11	7	8

Figure 13 - pgAdmin data output for the above query

▼ Abc Pivot	▼ # Pivot
Spend Category	Total Spend
Spent on Alcohol	189
Spent on Fish	111
Spent on Meat	379
Spent on chocolates	189
Spent on vegetables	104
Spent On Commodities	218
Spent on Alcohol	464

Figure 14 - a sample of the pivot table. The column head detailing pivot

Successful Channel

IF [Brochure ad] = 1 AND [Campaign Accept?] = 1 THEN "Brochure"
ELSEIF [Facebook ad] = 1 AND [Campaign Accept?] = 1 THEN "Facebook"
ELSEIF [Instagram ad] = 1 AND [Campaign Accept?] = 1 THEN "Instagram"
ELSEIF [Twitter ad] = 1 AND [Campaign Accept?] = 1 THEN "Twitter"
ELSEIF [Bulkmail ad] = 1 AND [Campaign Accept?] = 1 THEN "Bulkmail"
END

The calculation is valid.

5 Dependencies

Apply

OK

Figure 15 - a pop-up view of the calculated field using the IF function

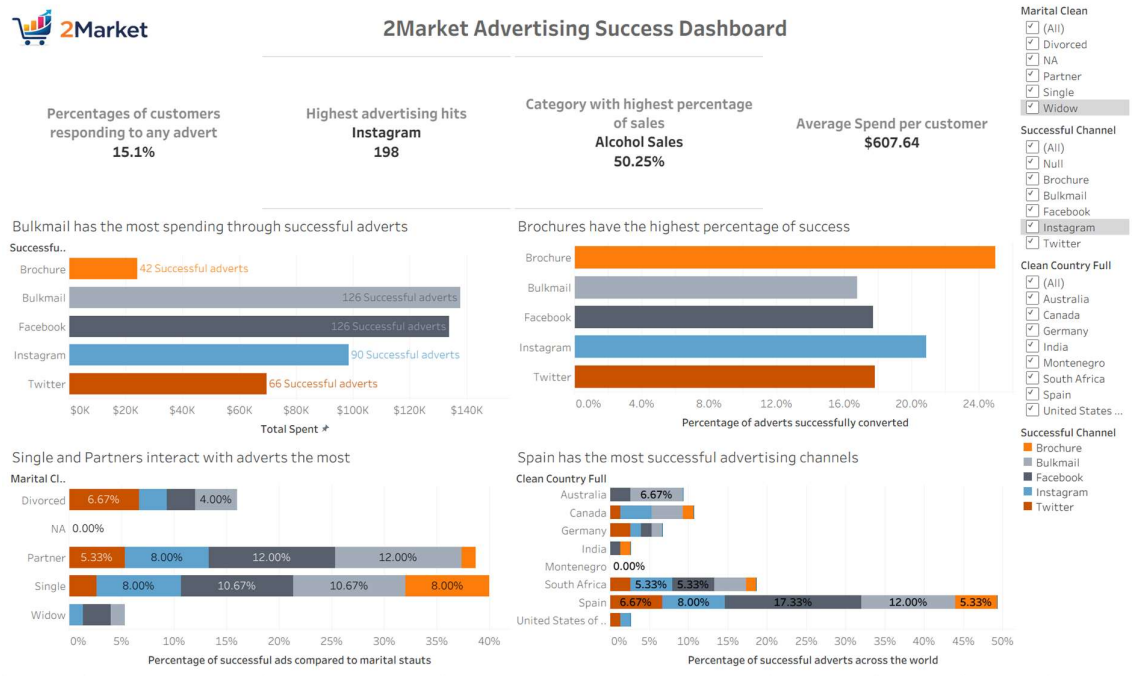


Figure 16 - The dashboard for the analysis